

Reference output

An Ordered Probit PMT Model on synthetic Susenas March 2022–2025 · the Stata output of the four do files

This is what the four completed do files print when they are run one after another on the delivered package. Your do files must reproduce every number here: every row count, every checkpoint table, every coefficient, every targeting error. Commands are not shown, only what they print. Step 2 holds the two linear models, step 3 the two ordered probits and, at the end, the inclusion and exclusion errors of all four models at the bottom 50 per cent.

Step 0 — s0_import.do: row counts of every imported file

March 2022

KOR22RT: 339584 rows
KOR22IND_1: 1237768 rows
blok43, 3 files appended: 339584 rows
blok42, 3 files appended: 825220 rows

March 2023

ssn202303_kor_rt: 341802 rows
ssn202303_kor_ind1: 1222683 rows
ssn202303_kp_blok43, 3 files appended: 341802 rows
ssn202303_kp_blok42, 3 files appended: 830079 rows

March 2024

ssn202403_kor_rt: 343479 rows
ssn202403_kor_ind1: 1210645 rows
ssn202403_kor_ind2: 1210645 rows
ssn202403_kp_blok43: 343479 rows
ssn202403_kp_blok42, 3 files appended: 842993 rows

March 2025

KOR25RT: 343460 rows
KOR25IND_1: 1174659 rows
KOR25IND_2: 1174659 rows
BLOK43, 3 files appended: 343460 rows
BLOK42, 3 files appended: 851210 rows

+-----+				
	year	r101	r105	povline
+-----+				
1.	2022	11	1	606394
2.	2022	11	2	565762
3.	2022	12	1	583221
4.	2022	12	2	534184
+-----+				

Step 0 complete. Imported files are in <your work folder>

Step 1 — s1_build.do: the analysis file of each round, and the deciles

Building March 2022

households in module 42: 339584

March 2022: 339584 households built

Variable	Mean	SD	Min	Max
-----+-----				
hhsiz	3.64	1.62	1.00	23.00
expend	4483708.88	3280778.81	279445.22	99286394.20
kwh	123.23	145.73	0.00	5951.00
credit	56953.10	82368.03	0.00	1321509.00
package	87391.69	122281.73	0.00	2449090.00
h_luaslantai	75.95	54.01	4.00	576.00
n_worker	1.69	0.90	0.00	9.00
hd_age	48.72	13.27	17.00	92.00
lnrpcexp	14.30	0.64	12.01	18.13
-----+-----				

Building March 2023

March 2023: 341802 households built

Variable	Mean	SD	Min	Max
-----+-----				
hhsiz	3.58	1.59	1.00	25.00
expend	4873308.32	3645134.82	330574.11	138585232.29
kwh	126.71	143.54	0.00	5284.20
credit	57144.36	85042.79	0.00	1314923.00
package	96956.73	128783.22	0.00	1411102.00
h_luaslantai	76.43	50.46	5.00	582.00
n_worker	1.71	0.90	0.00	9.00
hd_age	49.48	13.02	17.00	93.00
lnrpcexp	14.32	0.64	11.88	18.23
-----+-----				

Building March 2024

March 2024: 343479 households built

Variable	Mean	SD	Min	Max
-----+-----				
hhsiz	3.52	1.57	1.00	22.00
expend	5060359.21	3704901.59	342851.95	123936975.12
kwh	134.22	145.88	0.00	5201.80
credit	58433.43	85959.33	0.00	1426269.00
package	102239.02	128546.04	0.00	1722257.00
h_luaslantai	78.96	51.10	5.00	624.00
n_worker	1.72	0.89	0.00	10.00
hd_age	50.17	12.80	17.00	93.00
lnrpcexp	14.32	0.63	11.99	18.52
-----+-----				

Building March 2025

March 2025: 343460 households built

Variable	Mean	SD	Min	Max
-----+-----				
hhsiz	3.42	1.53	1.00	20.00
expend	5112432.66	3635463.26	357793.00	118041942.29
kwh	144.74	153.19	0.00	3097.10
-----+-----				

credit	55178.44	101022.47	0.00	8828917.00
package	111244.40	147064.69	0.00	4566881.00
h_luaslantai	78.53	51.32	6.00	500.00
n_worker	1.71	0.89	0.00	9.00
hd_age	49.82	13.05	16.00	94.00
lnrpcexp	14.32	0.61	11.64	18.29

10			
quantiles			
of rpcexp	Freq.	Percent	Cum.
1	83,362,517	10.00	10.00
2	83,364,558	10.00	20.00
3	83,362,041	10.00	30.00
4	83,361,372	10.00	40.00
5	83,361,984	10.00	50.00
6	83,364,294	10.00	60.00
7	83,360,806	10.00	70.00
8	83,362,060	10.00	80.00
9	83,363,086	10.00	90.00
10	83,361,736	10.00	100.00

Total | 833,624,454 100.00
training sample: 1024865 households

10			
quantiles			
of rpcexp	Freq.	Percent	Cum.
1	28,777,860	10.00	10.00
2	28,776,472	10.00	20.00
3	28,777,347	10.00	30.00
4	28,778,284	10.00	40.00
5	28,777,953	10.00	50.00
6	28,776,519	10.00	60.00
7	28,775,489	10.00	70.00
8	28,779,777	10.00	80.00
9	28,775,011	10.00	90.00
10	28,776,534	10.00	100.00

Total | 287,771,246 100.00
test sample: 343460 households

Step 1 complete.

		train	
year		0	1
			Total
2022	0	339,584	339,584
2023	0	341,802	341,802
2024	0	343,479	343,479
2025	343,460	0	343,460
Total	343,460	1,024,865	1,368,325
114,341			
114,341			
135,938			
62,763			

Step 2 — s2_pmt_ols.do: models 1 and 2, linear, district fixed effects

1. The PMT, district fixed effects

Variable	Active
ln_hhsize	-0.8860 0.0029
ln_hhsize_2	0.0598 0.0057
ln_h_luaslantai	0.0250 0.0062
ln_h_luaslantai_2	0.0133 0.0007
h_house	
2	-0.0163 0.0021
3	-0.0258 0.0015
4	0.0771 0.0038
5	-0.0229 0.0265
h_roof	
2	-0.0345 0.0035
3	-0.0366 0.0035
4	-0.0445 0.0038
5	-0.0246 0.0083
6	-0.0146 0.0072
7	-0.0433 0.0050
8	-0.0423 0.0087
h_wall	
2	-0.0127 0.0039
3	-0.0272 0.0011
4	-0.0607 0.0036
5	-0.0649 0.0072
6	-0.0531 0.0060
7	-0.0531 0.0035
h_floor	
2	-0.0950 0.0032
3	-0.1503 0.0075
4	-0.1451 0.0038
5	-0.1709 0.0035
6	-0.1675 0.0033
7	-0.2008 0.0061

8	-0.2430
	0.0040
9	-0.2422
	0.0086
h_toilet	
2	-0.0177
	0.0017
3	-0.0570
	0.0060
4	-0.0404
	0.0030
5	-0.0498
	0.0099
6	-0.0493
	0.0018
h_closet	
2	0.0405
	0.0047
3	0.0150
	0.0039
4	-0.0087
	0.0026
99	(omitted)
h_sewage	
2	0.0345
	0.0045
3	-0.0141
	0.0019
4	-0.0265
	0.0014
5	-0.0193
	0.0040
6	0.0121
	0.0081
99	(omitted)
h_water	
2	-0.1591
	0.0020
3	-0.1798
	0.0023
4	-0.2143
	0.0022
5	-0.2483
	0.0022
6	-0.2612
	0.0031
7	-0.2388
	0.0023
8	-0.2545
	0.0031
9	-0.2721
	0.0036
10	-0.2371
	0.0029
11	-0.2485
	0.0140
h_light	
2	0.0012
	0.0021
3	0.0538
	0.0025
4	-0.0485
	0.0034
h_fuel	

1	-0.0619
	0.0086
2	-0.0586
	0.0068
3	-0.0376
	0.0062
4	-0.1399
	0.0057
5	-0.0988
	0.0072
6	-0.1678
	0.0432
7	-0.1762
	0.0061
8	-0.0882
	0.1118
9	-0.2253
	0.0182
10	-0.2464
	0.0059
11	-0.1299
	0.0256
h_epower	
2	0.0601
	0.0010
3	0.1285
	0.0015
99	(omitted)
h_asset_a	0.0954
	0.0016
h_asset_b	0.0988
	0.0010
h_asset_c	0.1251
	0.0017
h_asset_d	0.1376
	0.0031
h_asset_e	0.1125
	0.0047
h_asset_f	0.1456
	0.0013
h_asset_g	0.1448
	0.0012
h_asset_h	0.1612
	0.0011
h_asset_i	0.0242
	0.0017
h_asset_j	0.0854
	0.0026
h_asset_k	0.2615
	0.0014
h_asset_l	0.0859
	0.0013
h_asset_m	0.0331
	0.0010
n_age04	0.0234
	0.0037
n_age519	0.0193
	0.0038
n_age2064	0.0224
	0.0037
n_age65up	0.0090
	0.0038
n_female	-0.0132
	0.0006
n_worker	0.0321
	0.0006
h_married	-0.0000
	0.0009
h_widowed	0.0010

	0.0013
h_neverschool	-0.0084
	0.0010
h_stillschool	-0.0005
	0.0010
head_female	-0.0736
	0.0016
hd_age	-0.0028
	0.0000
hd_edu	-0.0003
	0.0000
urban	-0.0694
	0.0013
_cons	14.7805
	0.0188

N	1024865
r2_a	0.5717
rmse	0.4175

Legend: b/se

2. The parsimonious model, district fixed effects

Variable	Active
ln_hhsize	-0.5597
	0.0024
ln_elec_kwh_pc	-0.0057
	0.0007
ln_telecom_cons_pc	-0.1419
	0.0003
ln_hhsize_2	0.0562
	0.0011
ln_elec_kwh_pc_2	0.0135
	0.0001
ln_telecom_cons_pc_2	0.0204
	0.0000
h_epower2	0.0294
	0.0030
h_epower3	0.0258
	0.0047
ln_elec_kwh_pc_h_epower2	0.0008
	0.0009
ln_elec_kwh_pc_h_epower3	0.0144
	0.0013
h_asset_car	0.2189
	0.0012
h_asset_computer	0.0902
	0.0011
h_asset_land	0.0276
	0.0009
ln_h_luaslantai	-0.0503
	0.0055
ln_h_luaslantai_2	0.0176
	0.0007
urban	-0.0571
	0.0010
_cons	13.5160
	0.0146

N	1024865
r2_a	0.6681
rmse	0.3675

Legend: b/se

Step 2 complete.

year	train		Total
	0	1	
2022	0	339,584	339,584
2023	0	341,802	341,802
2024	0	343,479	343,479
2025	343,460	0	343,460
Total	343,460	1,024,865	1,368,325

Step 3 — s3_pmt_ordered.do: models 3 and 4, ordered probit; and 5, the targeting errors

3. Ordered probit on the decile, the PMT, district fixed effects

	Variable	Active
decile	ln_hhsize	-2.0029 0.0077
	ln_hhsize_2	0.0302 0.0147
	ln_h_luaslantai	0.0743 0.0158
	ln_h_luaslantai_2	0.0306 0.0019
	h_house	
	2	-0.0388 0.0053
	3	-0.0639 0.0037
	4	0.1883 0.0095
	5	-0.0411 0.0666
	h_roof	
	2	-0.0779 0.0089
	3	-0.0847 0.0089
	4	-0.1011 0.0096
	5	-0.0588 0.0208
	6	-0.0417 0.0179
	7	-0.0989 0.0128
	8	-0.0886 0.0220
	h_wall	
	2	-0.0334 0.0098
	3	-0.0703 0.0028
	4	-0.1422 0.0089
	5	-0.1558 0.0181
	6	-0.1429 0.0151
	7	-0.1357 0.0087
	h_floor	
	2	-0.2217 0.0084
	3	-0.3583 0.0188
	4	-0.3446 0.0099
	5	-0.4130 0.0090
	6	-0.4019 0.0086

7	-0.4918
	0.0155
8	-0.5875
	0.0103
9	-0.5677
	0.0219
h_toilet	
2	-0.0420
	0.0044
3	-0.1210
	0.0152
4	-0.0964
	0.0075
5	-0.1158
	0.0248
6	-0.1214
	0.0045
h_closet	
2	0.1019
	0.0118
3	0.0345
	0.0098
4	-0.0208
	0.0065
99	(omitted)
h_sewage	
2	0.0647
	0.0114
3	-0.0334
	0.0048
4	-0.0651
	0.0035
5	-0.0472
	0.0101
6	0.0302
	0.0203
99	(omitted)
h_water	
2	-0.3625
	0.0053
3	-0.4137
	0.0059
4	-0.4956
	0.0057
5	-0.5778
	0.0058
6	-0.6132
	0.0078
7	-0.5563
	0.0060
8	-0.5960
	0.0079
9	-0.6464
	0.0092
10	-0.5553
	0.0075
11	-0.5776
	0.0350
h_light	
2	0.0026
	0.0051
3	0.1355
	0.0062
4	-0.1154
	0.0085

h_fuel	
1	-0.1574
	0.0221
2	-0.1523
	0.0178
3	-0.1000
	0.0162
4	-0.3445
	0.0151
5	-0.2509
	0.0187
6	-0.4297
	0.1096
7	-0.4397
	0.0159
8	-0.5383
	0.3036
9	-0.5606
	0.0456
10	-0.6137
	0.0155
11	-0.4001
	0.0657
h_epower	
2	0.1444
	0.0025
3	0.3029
	0.0038
99	(omitted)
h_asset_a	0.2284
	0.0040
h_asset_b	0.2408
	0.0025
h_asset_c	0.2966
	0.0043
h_asset_d	0.3207
	0.0081
h_asset_e	0.2463
	0.0124
h_asset_f	0.3511
	0.0032
h_asset_g	0.3498
	0.0030
h_asset_h	0.3913
	0.0028
h_asset_i	0.0587
	0.0044
h_asset_j	0.2071
	0.0066
h_asset_k	0.6276
	0.0035
h_asset_l	0.2057
	0.0032
h_asset_m	0.0823
	0.0026
n_age04	0.0928
	0.0097
n_age519	0.0801
	0.0098
n_age2064	0.0915
	0.0095
n_age65up	0.0574
	0.0098
n_female	-0.0319
	0.0016
n_worker	0.0784
	0.0015
h_married	0.0027

		0.0022
	h_widowed	0.0057
		0.0032
	h_neverschool	-0.0158
		0.0025
	h_stillschool	0.0002
		0.0024
	head_female	-0.1790
		0.0041
	hd_age	-0.0068
		0.0001
	hd_edu	-0.0007
		0.0001
	urban	-0.1667
		0.0031

	/cut1	-3.0038
		0.0478
	/cut2	-2.3948
		0.0478
	/cut3	-1.9414
		0.0478
	/cut4	-1.5483
		0.0478
	/cut5	-1.1733
		0.0478
	/cut6	-0.7924
		0.0478
	/cut7	-0.3773
		0.0478
	/cut8	0.1173
		0.0478
	/cut9	0.8142
		0.0478

Statistics		
	N	1024865
	r2_p	0.1700

Legend: b/se

log likelihood -1952894.7

Variable	Obs	Mean	Std. dev.	Min	Max

op_expected	343,460	6.153401	2.032667	1.05658	9.999998

4. Ordered probit on the decile, the parsimonious model, district fixed effects

Variable	Active

decile	
ln_hhsize	-1.5161
	0.0075
ln_elec_kwh_pc	-0.0172
	0.0020
ln_telecom_cons_pc	-0.3897
	0.0010
ln_hhsize_2	0.1397
	0.0034
ln_elec_kwh_pc_2	0.0374
	0.0004
ln_telecom_cons_pc_2	0.0561
	0.0001
h_epower2	0.0712
	0.0088
h_epower3	0.1061
	0.0143
ln_elec_kwh_pc_h_epower2	0.0048
	0.0026

ln_elec_kwh_pc_h_epower3	0.0293
	0.0039
h_asset_car	0.5970
	0.0035
h_asset_computer	0.2463
	0.0032
h_asset_land	0.0783
	0.0026
ln_h_luaslantai	-0.1146
	0.0159
ln_h_luaslantai_2	0.0460
	0.0019
urban	-0.1538
	0.0029

/cut1	-0.0541
	0.0422
/cut2	0.6250
	0.0422
/cut3	1.1332
	0.0423
/cut4	1.5748
	0.0423
/cut5	1.9977
	0.0423
/cut6	2.4289
	0.0423
/cut7	2.9009
	0.0423
/cut8	3.4671
	0.0423
/cut9	4.2726
	0.0424

Statistics	
N	1024865
r2_p	0.2200

Legend: b/se					
log likelihood	-1835298.3				
Variable	Obs	Mean	Std. dev.	Min	Max

ops_expected	343,460	6.191852	2.231353	1.006821	10

5. Targeting the bottom 50 per cent of the 2025 test round

IE = inclusion error, EE = exclusion error, both in per cent of people

model	IE50	EE50
1. PMT, linear, district FE	24.55	24.55
2. parsimonious, linear, district FE	21.05	21.05
3. PMT, ordered probit, district FE	24.52	24.53
4. parsimonious, ordered probit, district FE	21.09	21.09

Step 3 complete.